

LAB EXPERIMENT

ITA0506-COMPUTER VISION G

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EXPNO:5

AIM: Perform basic Image Handling and processing operations on the image

- **Read an image in python and Erode an Image using erode function.**

Algorithm:

- **Start and import libraries**
Start the program and import the required modules: cv2 for image processing and numpy for creating the structuring element (kernel).
- **Read the input image**
Read the image from the specified file path using cv2.imread() and store it in a variable.
- **Check if image is loaded**
Verify that the image is successfully loaded (not None). If loading fails, print an error message and terminate the program.
- **Convert image to grayscale**
Convert the color image from BGR format to grayscale using cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) to simplify further processing.
- **Define kernel and apply dilation**
Create a structuring element (for example, a 3×3 matrix of ones) using numpy, then apply the dilation operation on the grayscale image using cv2.dilate(gray, kernel, iterations=1) to thicken bright regions.
- **Display results and exit**
Display the original, grayscale, and dilated images using cv2.imshow(), wait for a key press using cv2.waitKey(0), and finally close all display windows using cv2.destroyAllWindows() to end the program.

Code:

```
EXP-5.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experime...
File Edit Format Run Options Window Help

import cv2
import numpy as np

# Read the image
img = cv2.imread("MICKEY.png")

# Check whether image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Create a 5x5 kernel
    kernel = np.ones((5,5), np.uint8)

    # Erode the image
    eroded = cv2.erode(img, kernel, iterations=1)

    # Display images
    cv2.imshow("Original Image", img)
    cv2.imshow("Eroded Image", eroded)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help

Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-5.py
Error: Image not found!

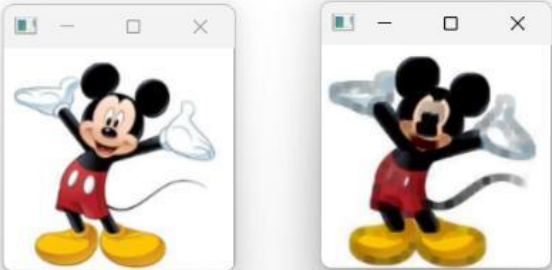
>>> = RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-5.py

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= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-5.py

Ln: 1 Col: 0
```



Result

The image was successfully read and the erosion operation was performed using the `cv2.erode()` function. The output image shows the foreground objects reduced in size due to erosion.